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@PuckerLab

Prof. Dr. Boas Pucker

Long-Read Plant Genomics

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Availability of slides

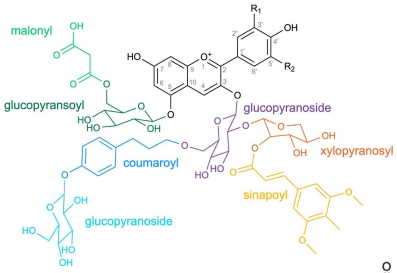
- All materials are freely available (CC BY) - after the lectures:
  - GitHub: https://github.com/bpucker/LRG

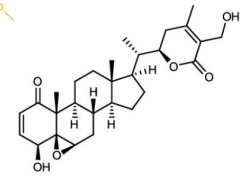
- Questions: Feel free to ask at any time

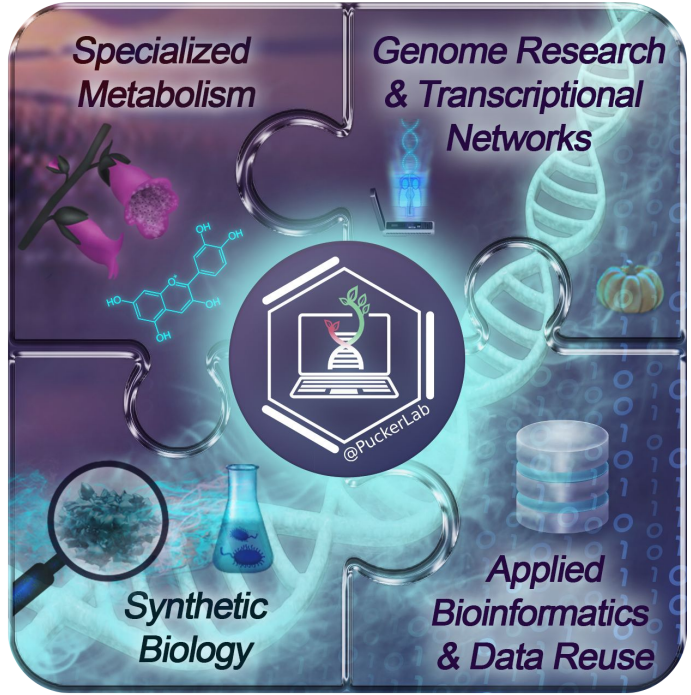
- Feedback, comments, or questions: pucker[a]uni-bonn.de

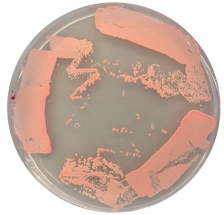
My figures and content can be re-used in accordance with CC BY 4.0, but this might not apply to all images/logos.

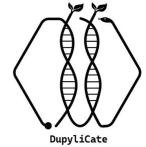
Plant Biotechnology and Bioinformatics











Wolff et al., 2023: 10.1515/jib-2023-0033
Dassow et al., 2025: 10.1101/2025.01.31.635922
Grünig et al., 2025: 10.1186/s12870-025-08006-3

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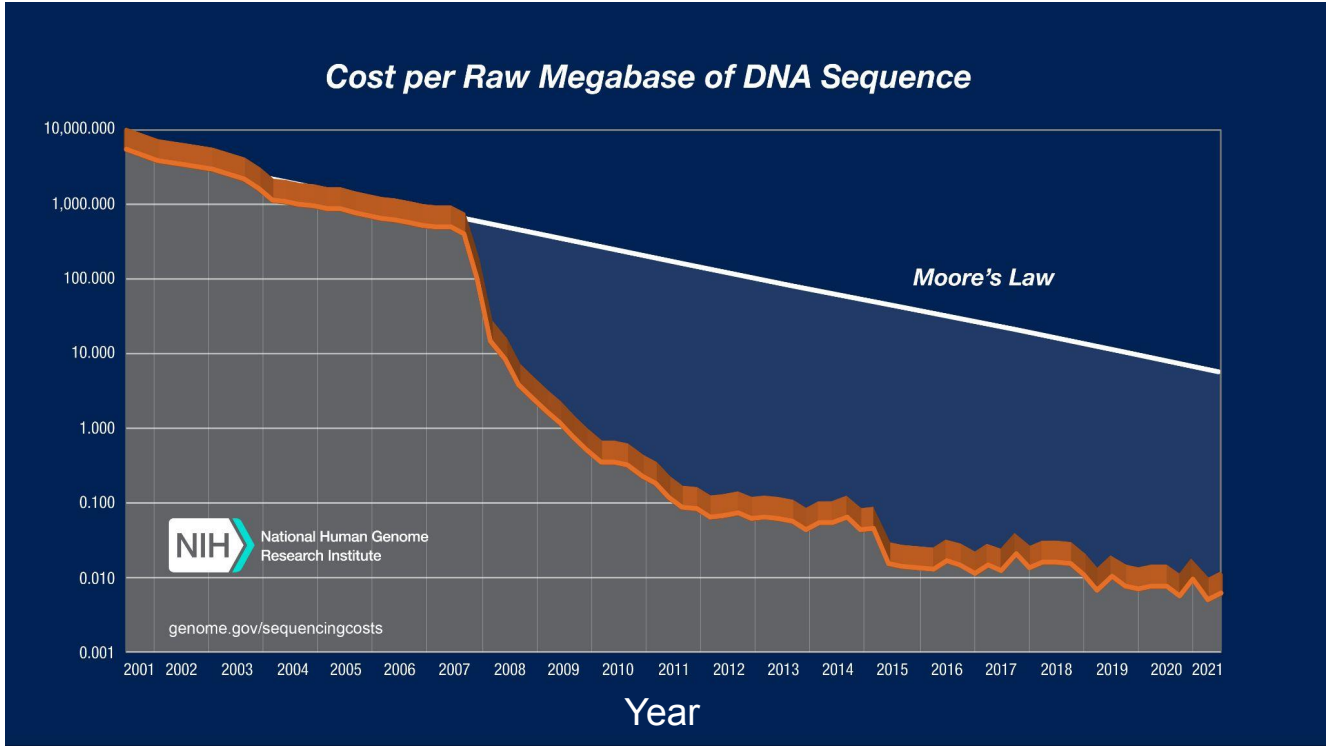
Development of sequencing capacity



https://www.genome.gov/about-genomics/fact-sheets/DNA-Sequencing-Costs-Data

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Sequencing technologies (selection)

- Generation 1:
  - **Sanger sequencing**
  - Max-Gilbert sequencing
- Generation 2 (massive parallel sequencing):
  - 454 pyrosequencing
  - Solexa/**Illumina sequencing**
  - SOLID
  - Ion Torrent
  - BGI-seq
  - Synthetic long reads
- Generation 3 (long reads):
  - **Pacific Biosciences (PacBio)**
  - **Oxford Nanopore Technologies (ONT)**
- Generation 4:
  - What is next?

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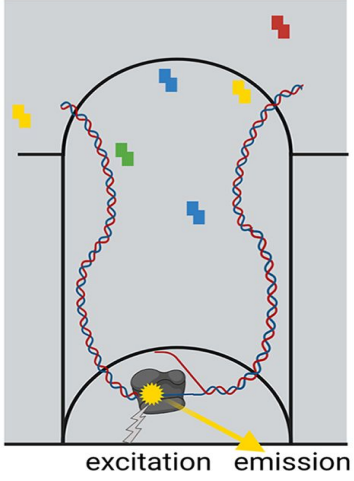
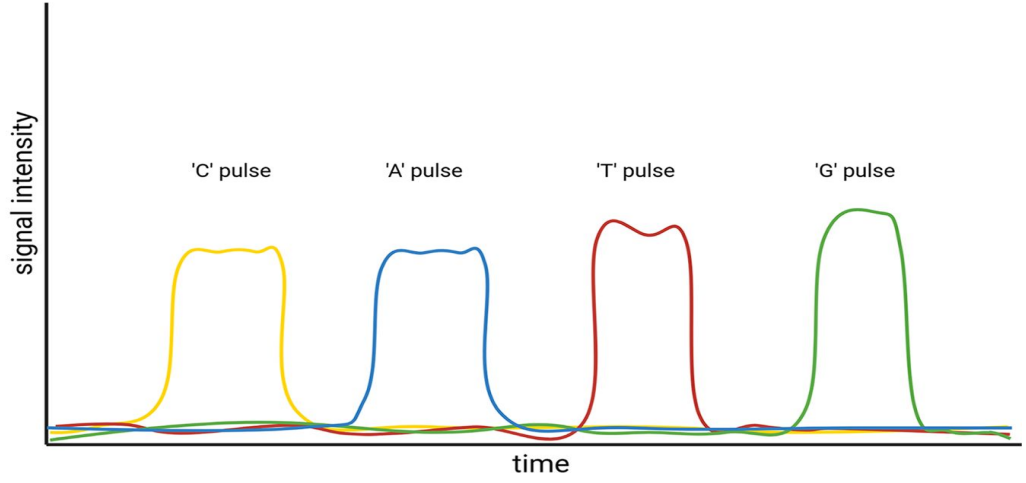
PacBio

Pacific Biosciences (PacBio)

- Polymerase located at bottom of well

- ZMWs = Zero Mode Waveguide





Pucker et al., 2022: 10.1017/qpb.2021.18

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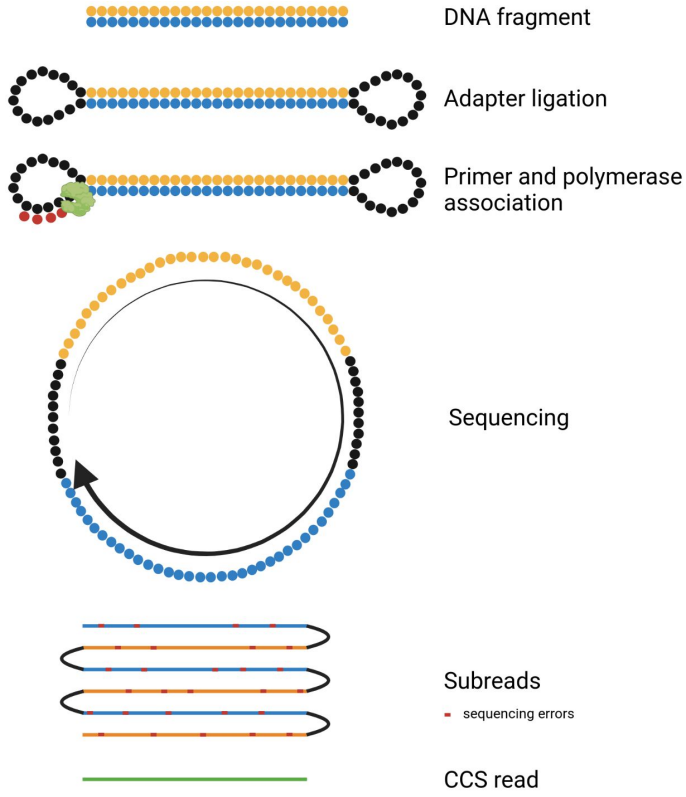
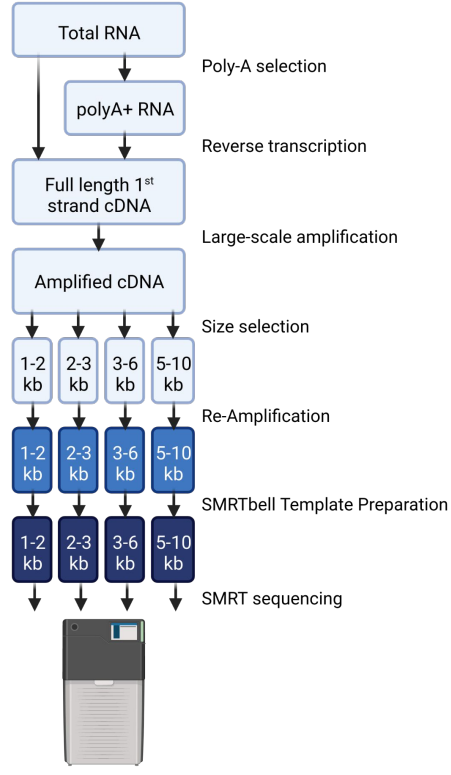
PacBio - HiFi



PacBio Iso-Seq (cDNA sequencing)

- SMRT = Single Molecule Real Time sequencing
- Full length cDNA sequencing is beneficial for gene prediction
- Iso-Seq generates several kb long reads and not only 2x300bp



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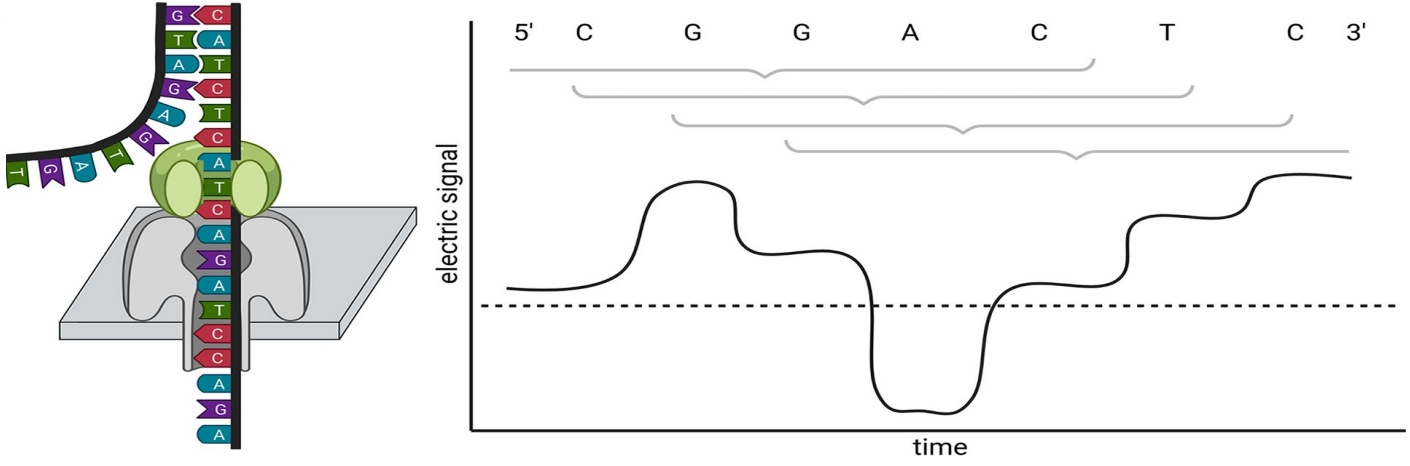
ONT

Oxford Nanopore Technologies (ONT)

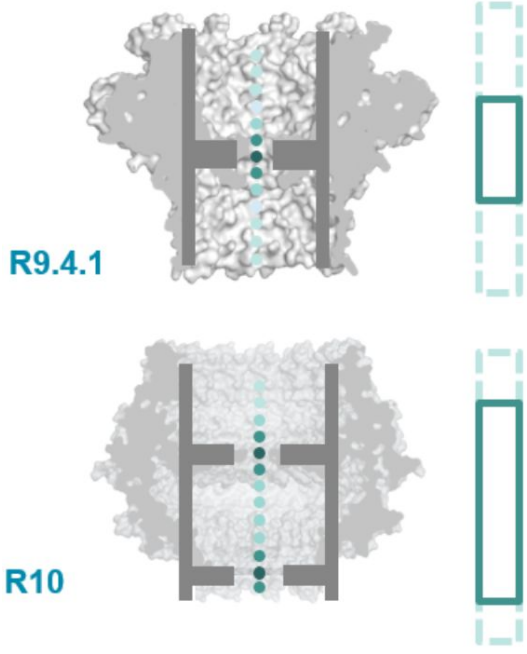
Under development since the 1980s

“Analysis of anything, anywhere by anyone” … not restricted to DNA sequencing



Pucker et al., 2022: 10.1017/qpb.2021.18

Nanopores



https://nanoporetech.com/about-us/news/r103-newest-nanopore-high-accuracy-nanopore-sequencing-now-available-store

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Plant genome sequencing workflow

How to sequence a plant genome?

Fresh plant material without legal restrictions

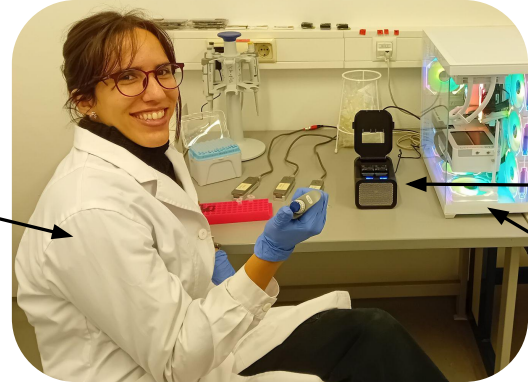
Plant DNA extraction and preparation for ONT sequencing

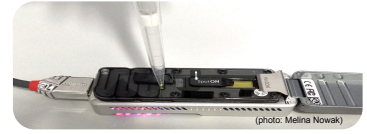
DOI
dx.doi.org/10.17504/protocols.io.bcvyiw7w
Boas Pucker[1]
[1]Plant Biotechnology and Bioinformatics, Institute of Plant Biology, TU Braunschweig

MAR 21, 2020

Established DNA extraction protocol

Highly skilled personnel





High tech devices

Extensive practice

de.NBI
GERMAN NETWORK FOR BIOINFORMATICS INFRASTRUCTURE

Cloud computing resources

MinKNOW

Sophisticated software

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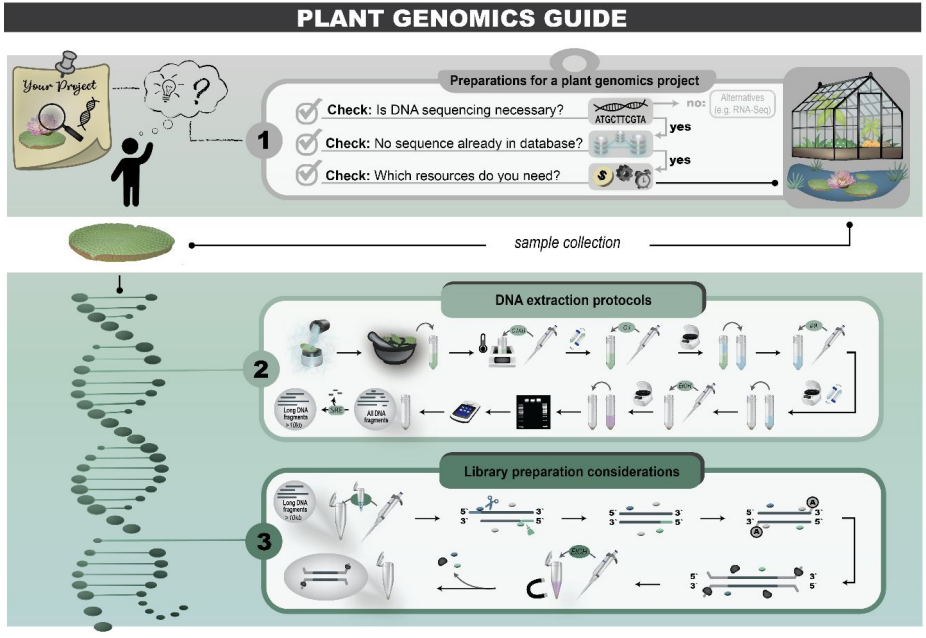
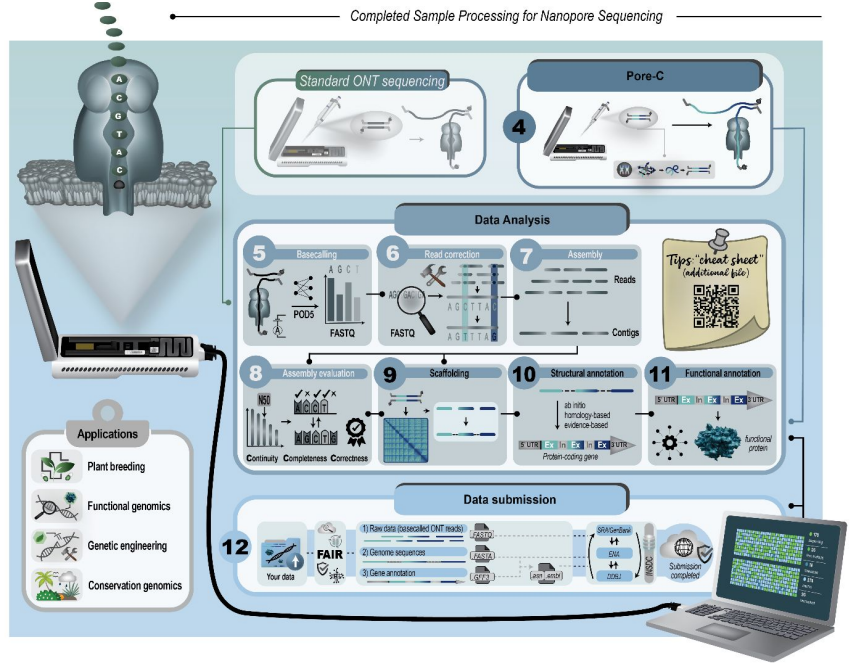
Plant Genomics Cookbook





Supplementary file with all commands!

de Oliveira et al., 2026: 10.1186/s12864-026-12623-z

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ONT sequencing workflow

		task	consumed time	hands-on time	equipment	estimated costs of consumables	estimated costs of lab equipment
A		plant incubation in darkness	2-3d	1h			
B		non-destructive sampling	-	1h			
C		DNA extraction	1d	8h	waterbath, centrifuge	$50	$1000 $8000
D		quality control	1h	1h	NanoDrop, Qubit	$20	
E		short fragment depletion	2h	1h	centrifuge	$50	
F		quality control	1h	1h	NanoDrop, Qubit	$20	$5000 $5000
G		library preparation & sequencing	1-5d	4-16h	centrifuge, magnetic rack, sequencer	$3000	$250 $1000
H		basecalling	1d	1h	computer with GPU		$3000
I		assembly	1-15d	1h			
J		polishing	1-5d	1h	compute cluster / cloud		
K		annotation	1-5d	1h			
L		data submission	2h	2h	fast internet connection		

Pucker et al., 2022: 10.1017/qpb.2021.18

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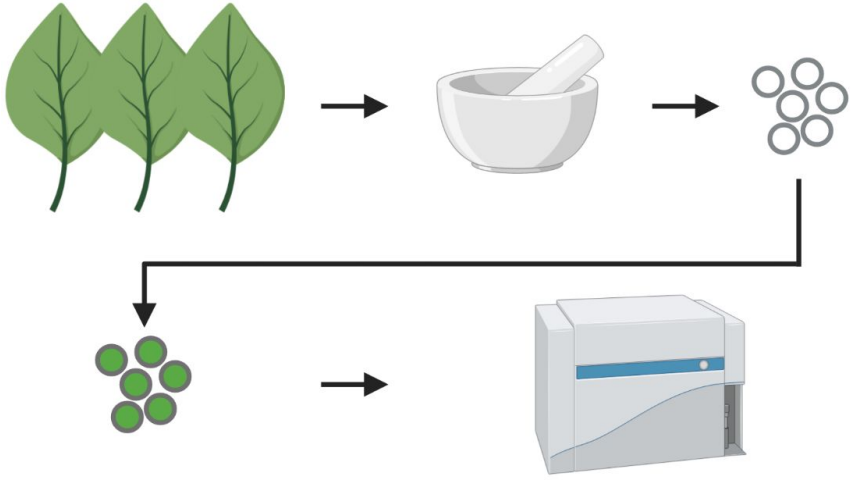
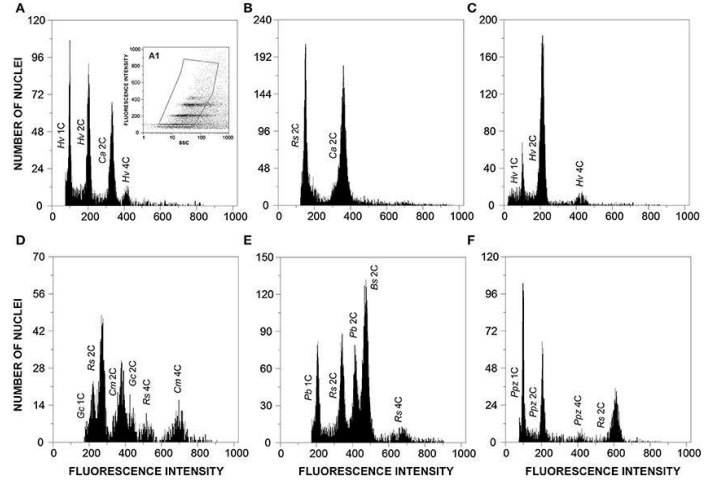
Considerations & preparations

- What is the expected/estimated genome size?
- What is the ploidy of the species?
- Which individual should be sequenced?
- Which plant parts are suitable for DNA extraction?
- What materials are needed for DNA extraction and sequencing?

Genome size

- Genome sequences of closely related species
- Flow cytometry is used to measure genome size biochemically
  - C-value
- Databases for plant genome sizes: https://cvalues.science.kew.org/

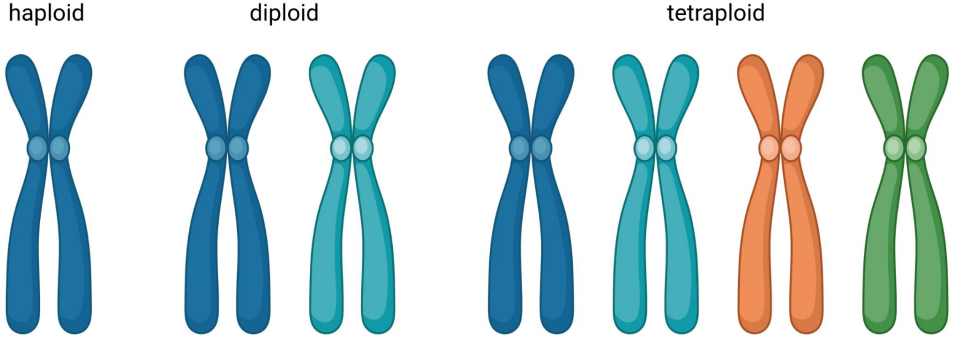


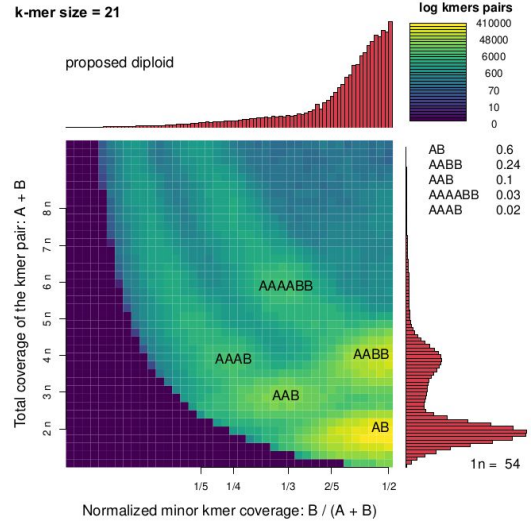


https://doi.org/10.3389/fpls.2014.00422

Ploidy

- Ploidy = copy number of same chromosome
- Many plants are polyploid
- Some polyploid plants have close diploid relatives
- Smudgeplots to analyze polyploidy based on short reads





Siadjeu & Pucker et al., 2020: 10.3390/genes11030274

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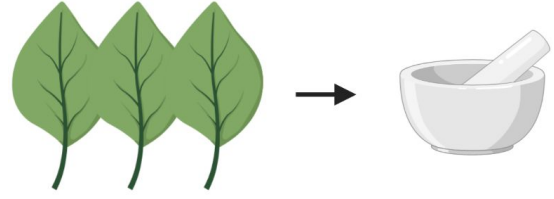
Picking an individual

- Plant should survive the sampling for DNA extraction
- Plant should be a good representation of the species
- Source of the plant is important:
  - Restrictions through material transfer agreements (MTAs)?
  - Restrictions through Nagoya protocol / Access and Benefit Sharing (ABS) law

Plant part for DNA extraction

- Young leaf is often a good choice
- Small cells result in higher density of nuclei per weight
- Concentration of specialized metabolites should be low
- Amount of sugar should be low
- Amount of chloroplast should be low
- Sample should not be contaminated with bacteria/fungi



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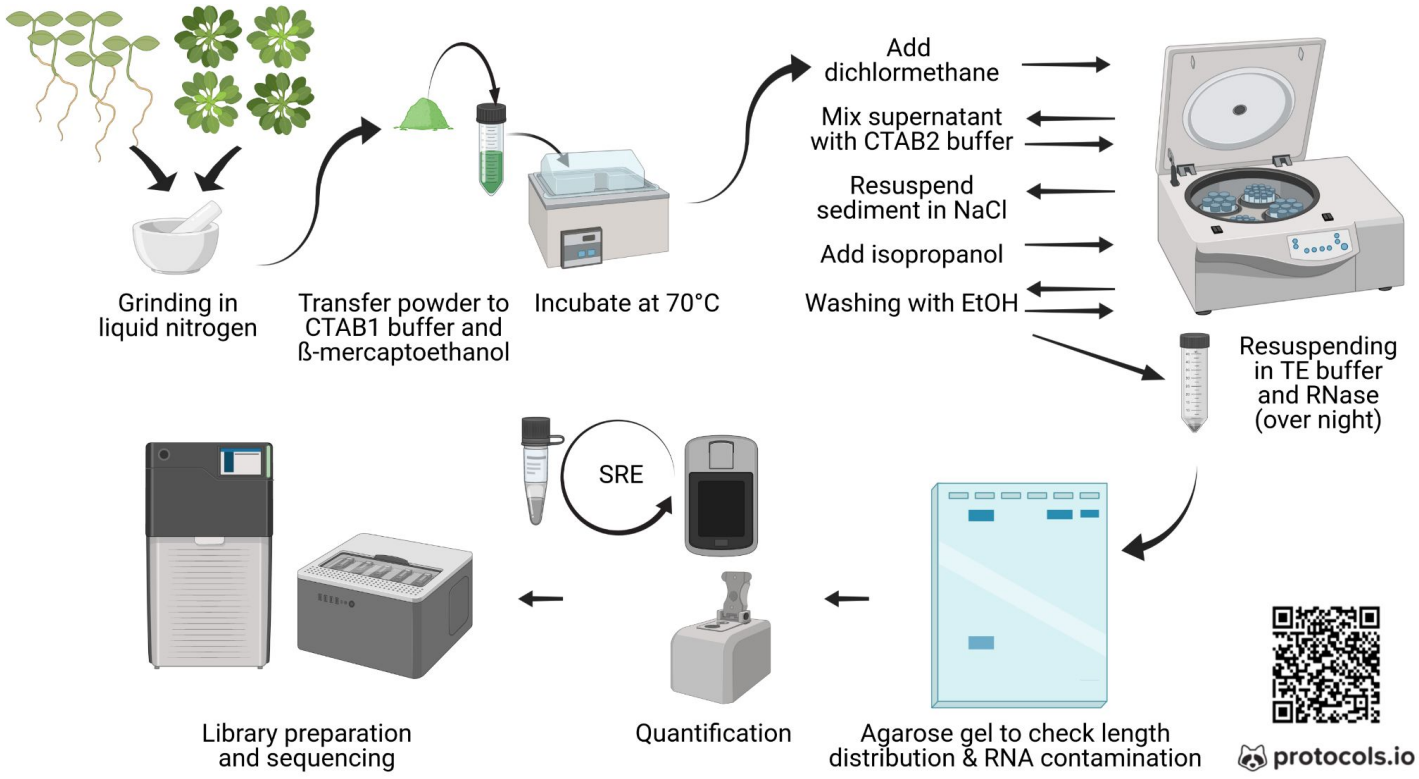
DNA extraction workflow



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DNA extraction challenges

- High amount of cpDNA is a big issue in sequencing projects
  - 50-100 chloroplasts per cell with 50-100 plastomes per chloroplast

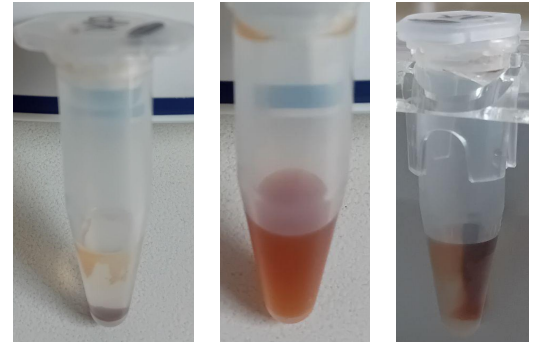
- Sequencing capacity is wasted on the cpDNA molecules
  - Very high coverage of the plastome; reduced coverage of nucleome

- Reducing amount of chloroplasts by incubating plants in the dark for some days prior to DNA isolation
  - Reduced amount of chloroplasts
  - Reduced concentration of starch/sugar

Golczyk et al., 2014: 10.1105/tpc.113.117465

Material required for sequencing

- DNA repair (companion module)
- Magnetic beads for purification
- Library preparation kit
- Sequencing kit
- Wash kit with DNase



ONT sequencing devices

MinION

PromethION 2 Solo

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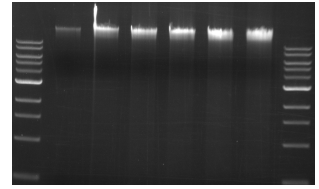
Flow cells: MinION vs. PromethION

Quality control

- Agarose gel electrophoresis

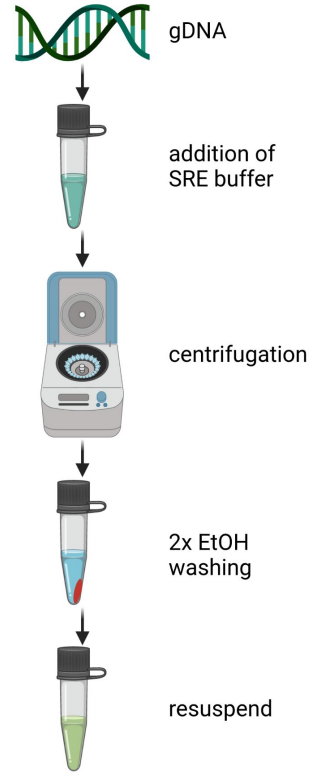


- Photometric measurement via NanoDrop

- Quantification with Qubit

Short Read Eliminator (SRE)

- Proprietary salt mix for DNA precipitation
- Removal of <10kb DNA fragments
- Depletion of <20kb DNA fragments
- ONT read length distribution can be substantially improved



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DNA repair

- Repairing single strand DNA breaks

- Repairing DNA ends (3’-A overhang required for adapter ligation)

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Library preparations

- Repaired DNA is subjected to library preparation
- Addition of adapters to DNA fragments (expensive ligase)
- Concentration of DNA can be quantified via Qubit measurement (optional)
  - Control step to ensure that library construction is working
- Purification of DNA with magnetic beads

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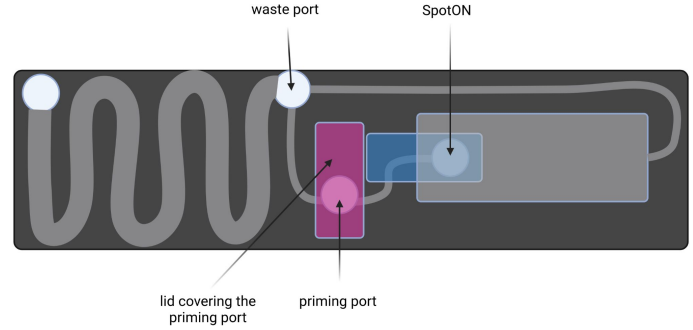
Flow cell check

- Flow cells are delivered with storage buffer (green)
- Buffer allows technical check of flow cell (number of active pores is determined)
- Number of nanopores must be >800 (MinION) or >5000 (PromethION)
- Replacement flow cells are provided if number of pores is lower

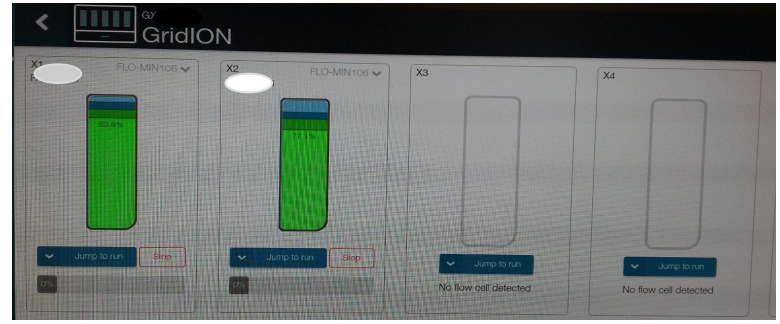
Loading flow cell

- Removal of storage buffer
- Priming of flow cell
- Introduction of air bubbles must be avoided!!!
- Fully open ports are crucial to inject solutions (avoid force)
- Video tutorial: https://www.youtube.com/watch?v=Pt-iaemrM88



Monitor sequencing

- Number of active nanopores can be monitored in real time
- Output is estimated in real time
- Read length distribution can be assessed
- Speed of the sequencing can be monitored
- Quality of the reads is displayed



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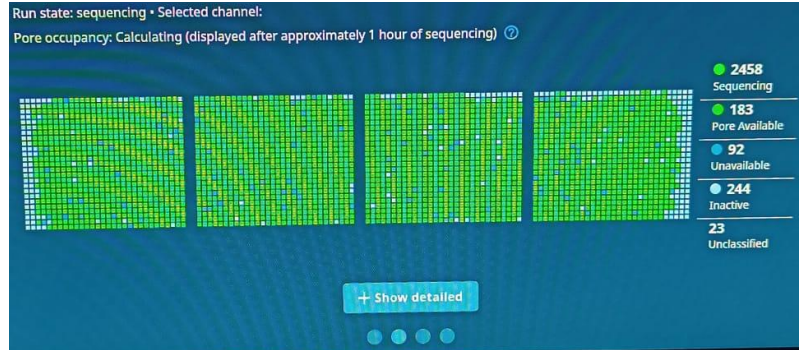
Nanopore activity

- Status of nanopores on flow cell
- Light green is good (sequencing)
- Number of sequencing nanopores change over time
- Air bubbles can destroy nanopores



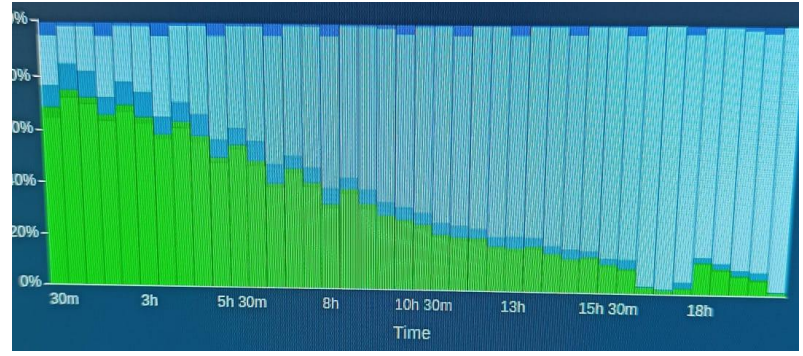
Read length distribution

- Read length distribution

- N50 summarizes length distribution

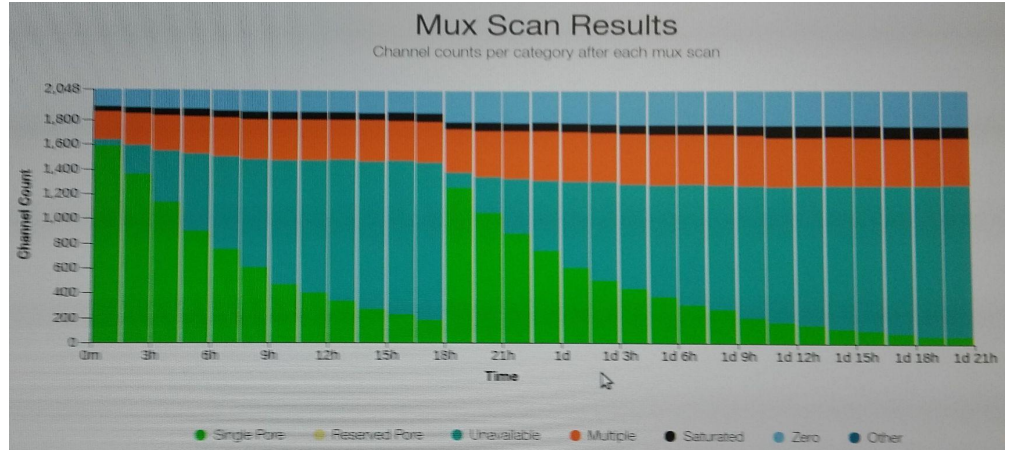
Flow cell performance over time

- Flow cell performance decreases over time
- Nanopores get blocked by DNA
- Nanopores break down due to air bubbles



Stop & wash

- Sequencing is stopped once the number of active pores gets low
- Washing with DNase to free blocked nanopores
- Flow cells are regenerated and can be re-used
- Process can be repeated multiple times (3-5x)



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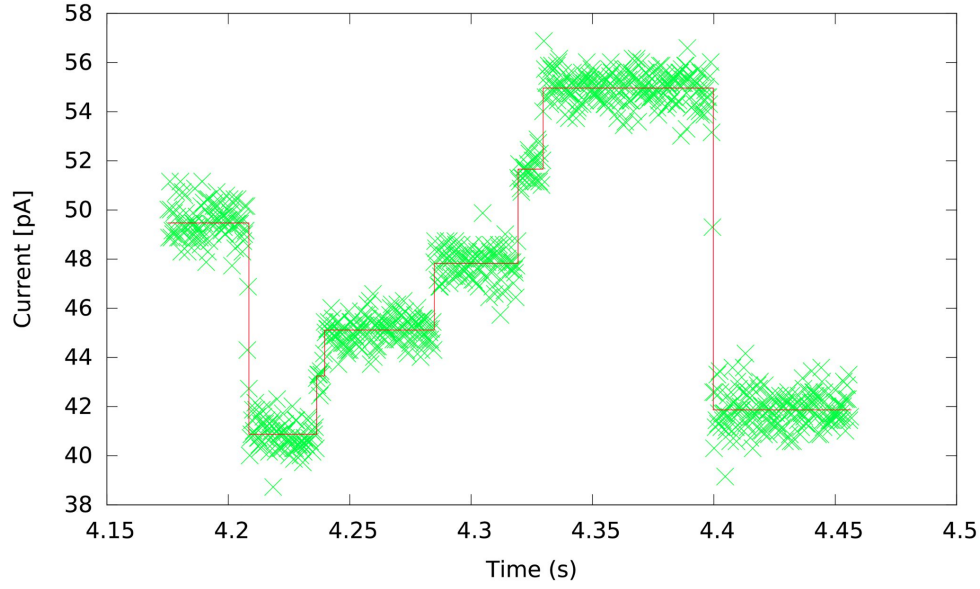
Basecalling

- Electric signal is converted into sequence information (bascalling)
- Algorithmic improvement lead to higher read accuracy
- Raw sequencing data need to be stored



Boza et al., 2017: 10.1371/journal.pone.0178751

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Basecalling with dorado

- Dorado allows conversion of electric signal into sequences
- Different models are available (HAC, SUP)
- Basecalling on CPUs is basically impossible
- Time estimates on GPUs: hours to days depending on dataset size

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Storing sequencing data in FASTQ

- Standard format for sequencing data
- Four lines per entry:
  - Header starts with @ (title + description)
  - Sequence
  - + (optional repetition of header)
  - Quality (phred encoded in ASCII character)
- Different versions existed (outdated today)
- Example:

@seq1

ACGTACGTACGT

+

“”?CB””:DC”

Symbol	Phred Quality Score	Probability of Incorrect Base Call
!	0	1.000
"	1	0.794
#	2	0.631
$	3	0.501
%	4	0.398
&	5	0.316

Phred quality score	Error probability	Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%

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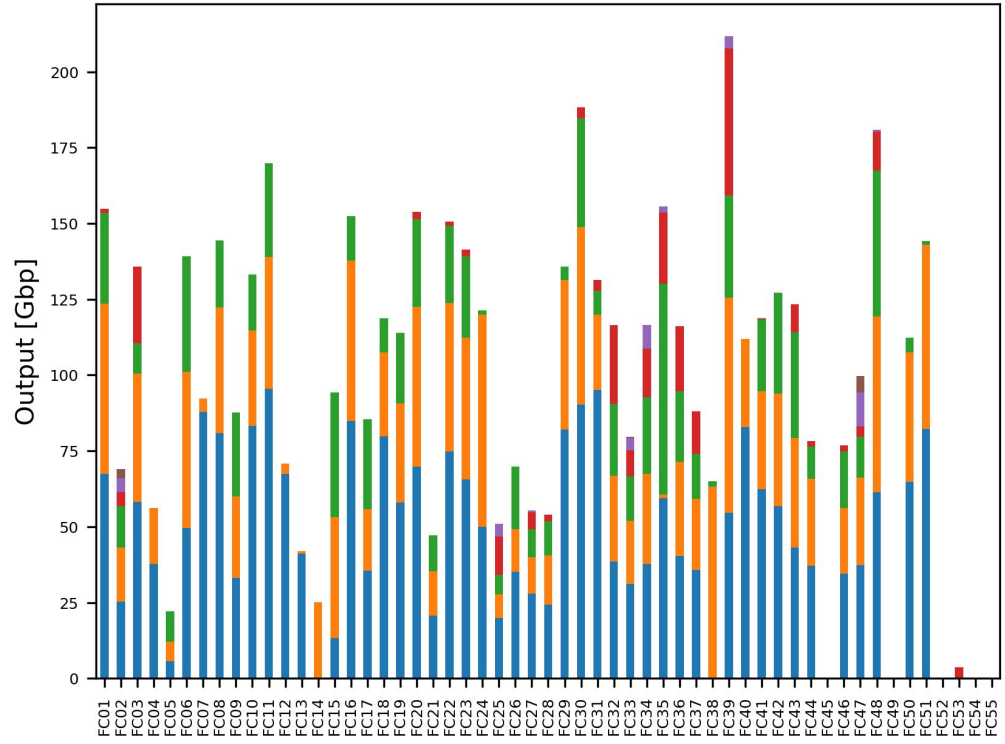
Calculating statistics

- Huge variation in output per flow cell
- Best runs: 200 Gbp
- Worst runs: 20 Gbp
- N50 for plants usually 25-65kb



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Genome Sequence Assembly

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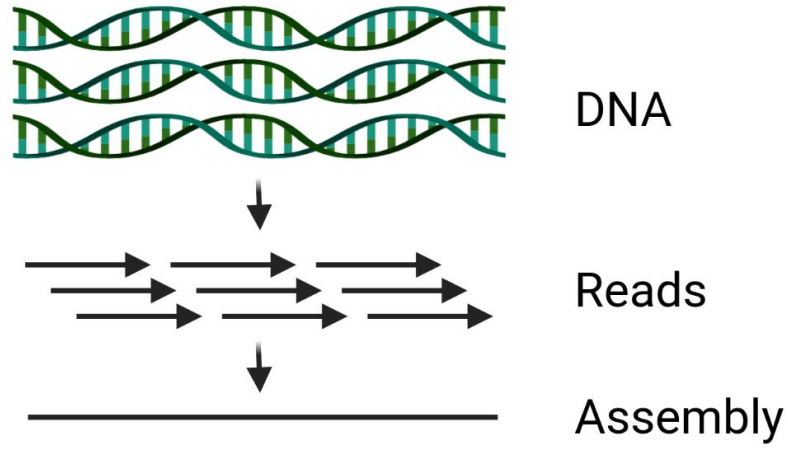
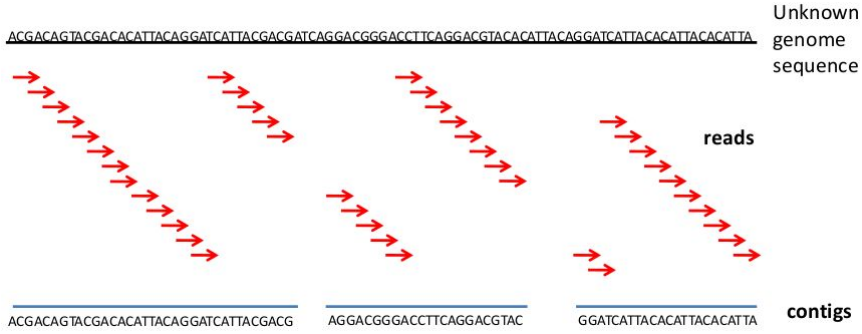
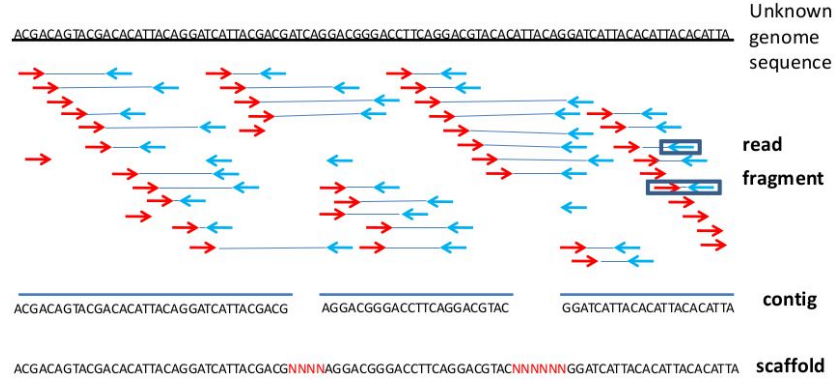
The assembly problem

- Reads are shorter than the chromosome - even long reads
- Multiple copies of the genome (DNA exist that can be subjected to sequencing
- Assembly = putting sequence pieces together (finding the common string of all substrings)
- Genome = DNA in a cell
- Genome sequence = representation of the DNA in a cell



Contigs, scaffolds, pseudochromosomes





Pseudochromosomes = scaffolds representing an entire chromosome
Gaps = regions between contigs that are represented by Ns

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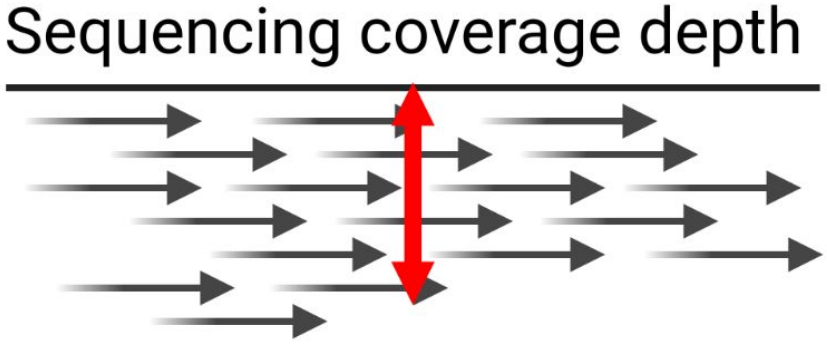
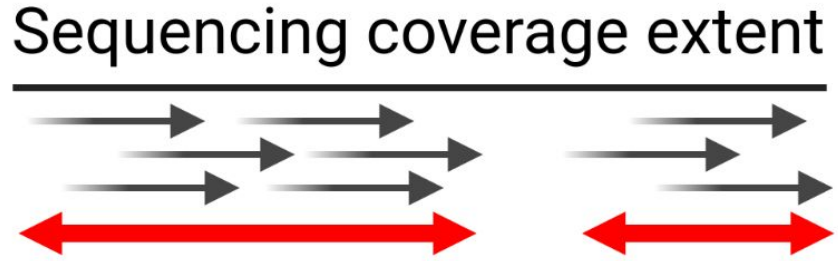
Sequencing coverage depth vs. coverage extent

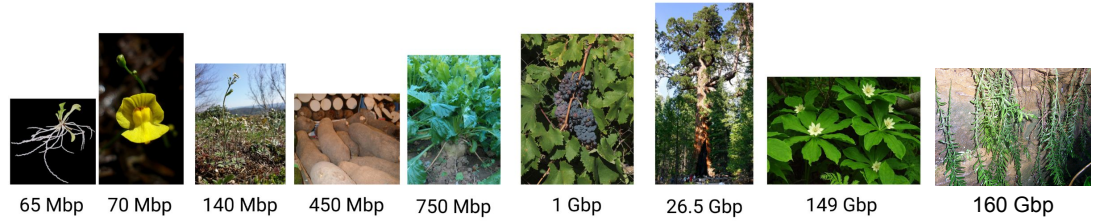




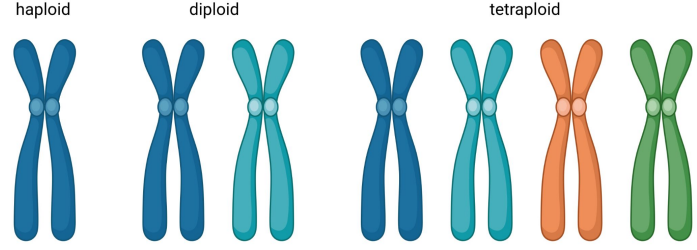
Assembly challenges

- Genome size: variation from 65 Mbp to 160 Gbp

- Ploidy: haploid/diploid genomes are much easier to analyze than polyploid genomes



Genlisea violacea: https://commons.wikimedia.org/wiki/File:Genlisea_violacea_giant.jpg
Utricularia gibba: https://commons.wikimedia.org/wiki/File:Utricularia_gibba_flower_01.jpg
Arabidopsis thaliana: https://commons.wikimedia.org/wiki/File:Arabidopsis_thaliana_sl10.jpg
Dioscorea dumetorum: https://commons.wikimedia.org/wiki/File:YamsatBrixtonMarket.jpg
Beta vulgaris: https://de.wikipedia.org/wiki/Zuckerr%C3%BCbe#/media/Datei:Zuckerr%C3%BCbe.jpg
Vitis vinifera: https://commons.wikimedia.org/wiki/File:Vitis_vinifera_20120823.jpg
Sequoia giganteum: https://commons.wikimedia.org/wiki/File:Grizzly_Giant_Mariposa_Grove.jpg
Paris japonica: https://commons.wikimedia.org/wiki/File:Paris_japonica_Kinugasasou_in_Hakusan_2003_7_27.jpg
Tmesipteris truncata: https://en.wikipedia.org/wiki/Tmesipteris_truncata#/media/File:Tmesipteris_truncata_Turramurra.jpg, CC BY-SA 3.0



Kress et al., 2022: 10.1073/pnas.2115640118
10.1038/d41586-024-01567-7

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Assembly evaluation

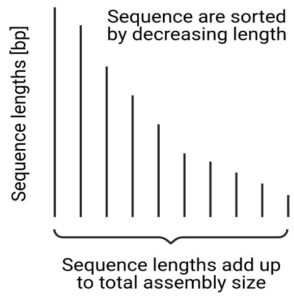
- Continuity: Does the assembly represent a genome in a small number of contigs?
- Completeness: Are all parts of the genome represented?
- Correctness: Is the assembly a correct representation of the genome?

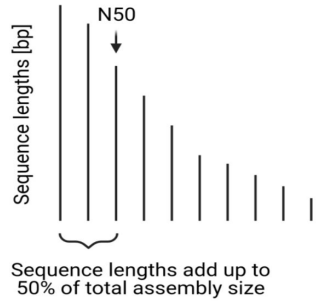
Evaluation: continuity

- Check assembly continuity: calculation based on sequences in FASTA file

- Number of contigs, assembly size, N50

assembler	Canu	FALCON	Miniasm	Flye
number of contigs	69	26	72	44
assembly size	123.5 Mbp	119.5 Mbp	120.2 Mbp	117 Mbp
maximal contig length	15.9 Mbp	15.9 Mbp	14.3 Mbp	14.9 Mbp
N50	13.4 Mbp	9.3 Mbp	8.6 Mbp	10.6 Mbp
N90	2.9 Mbp	2.8 Mbp	1.4 Mbp	2.5 Mbp





Pucker et al., 2019: 10.1371/journal.pone.0216233

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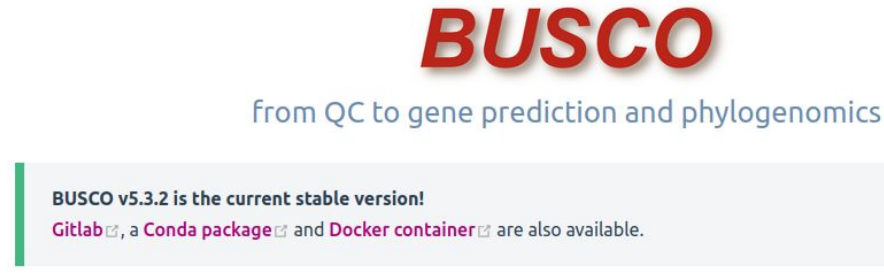
Evaluation - completeness

- Check assembly completeness: inspection for presence of conserved genes
- BUSCO = Benchmarking Universal Single-Copy Orthologue
- BUSCO genes are used to assess assembly completeness



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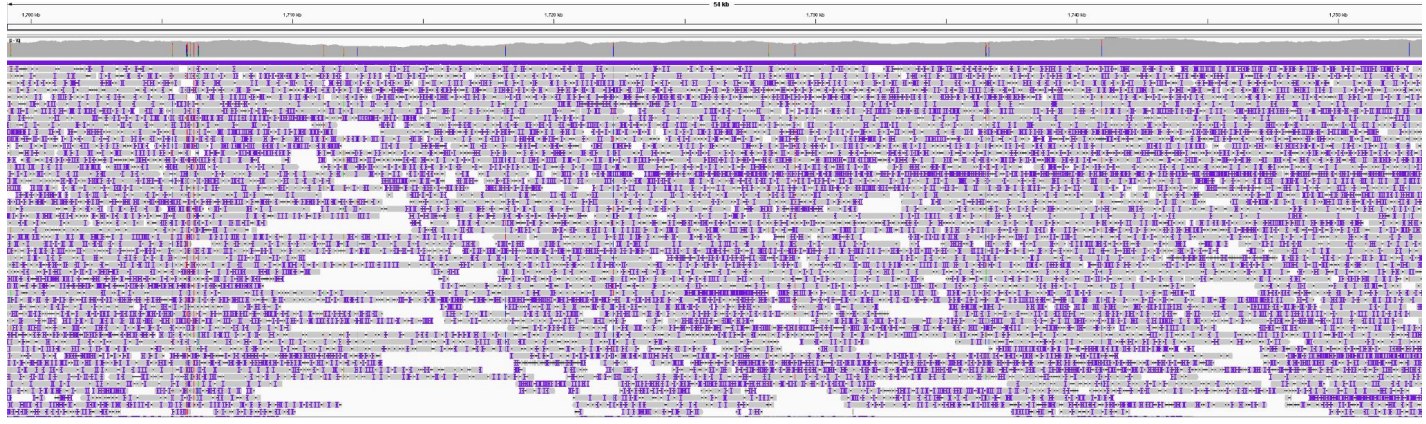
Evaluation - assembly correctness

- Check assembly correctness: analyses of read mappings
- Integrative Genomics Viewer (IGV) can visualize read mappings
- Tools: REAPR, SQUAT



IGV: Thorvaldsdottir et al., 2013: 10.1093/bib/bbs017
REAPR: Hunt et al., 2013: 10.1186/gb-2013-14-5-r47
SQUAT: Yang et al., 2019: 10.1186/s12864-019-5445-3

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Important assembler features

- NextDenovo2: works well with R9 reads
- Verkko: can integrate Pore-C data for scaffolding
- Shasta: super fast and resource-efficient
- Hifiasm: excellent assembler for highly accurate long reads

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Structural annotation

Finding genes in a genome sequence

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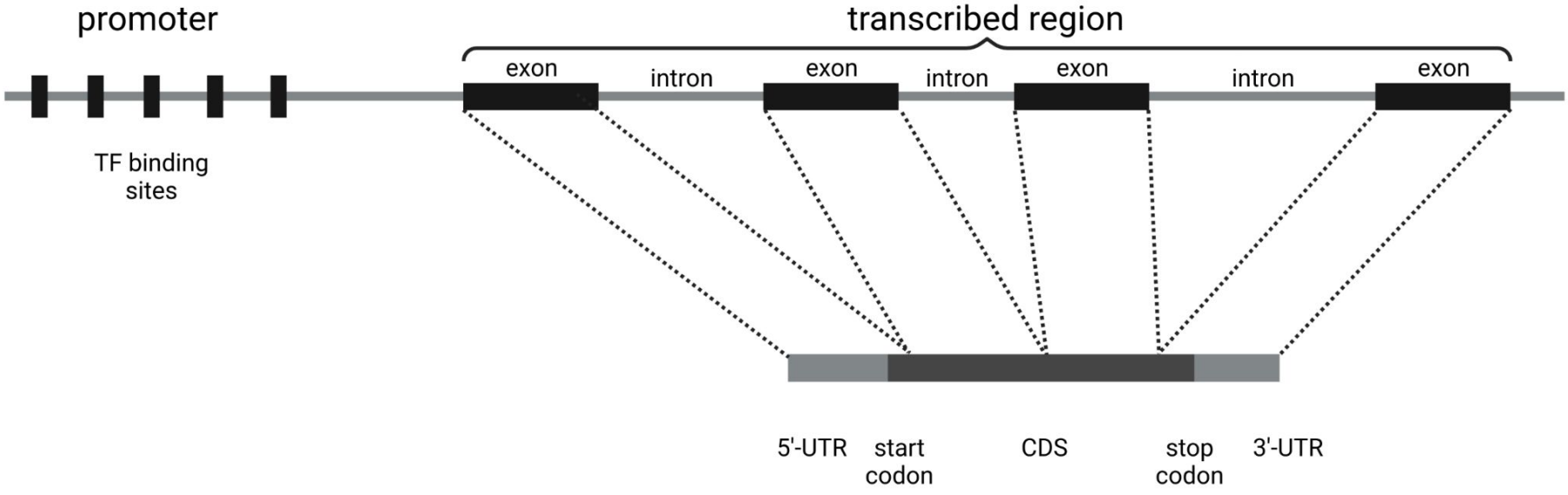
Siadjeu & Pucker et al., 2020: 10.3390/genes11030274

Plant gene structure

CDS = (Protein) Coding Sequence

ORF = Open Reading Frame



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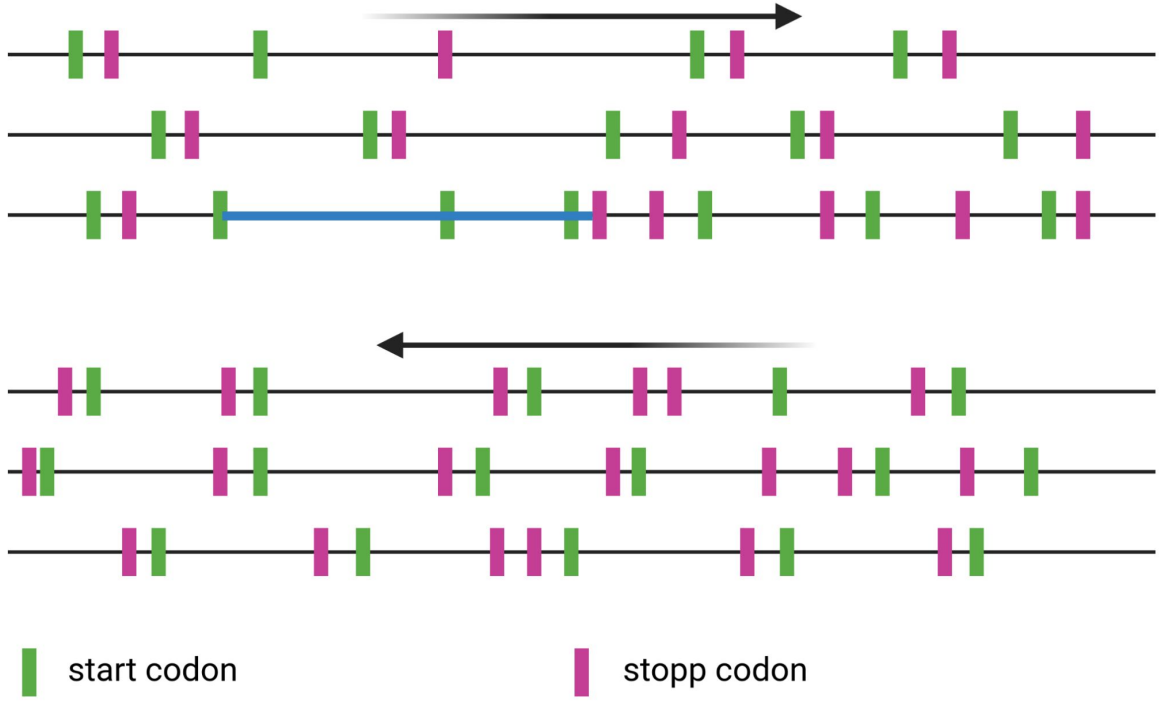
Finding ORFs



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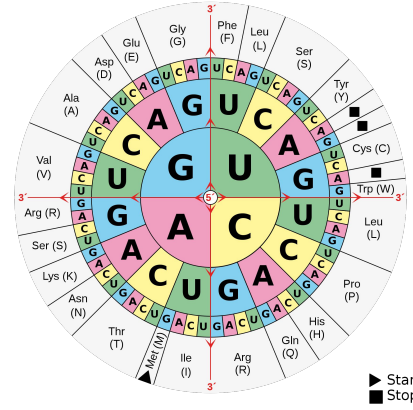
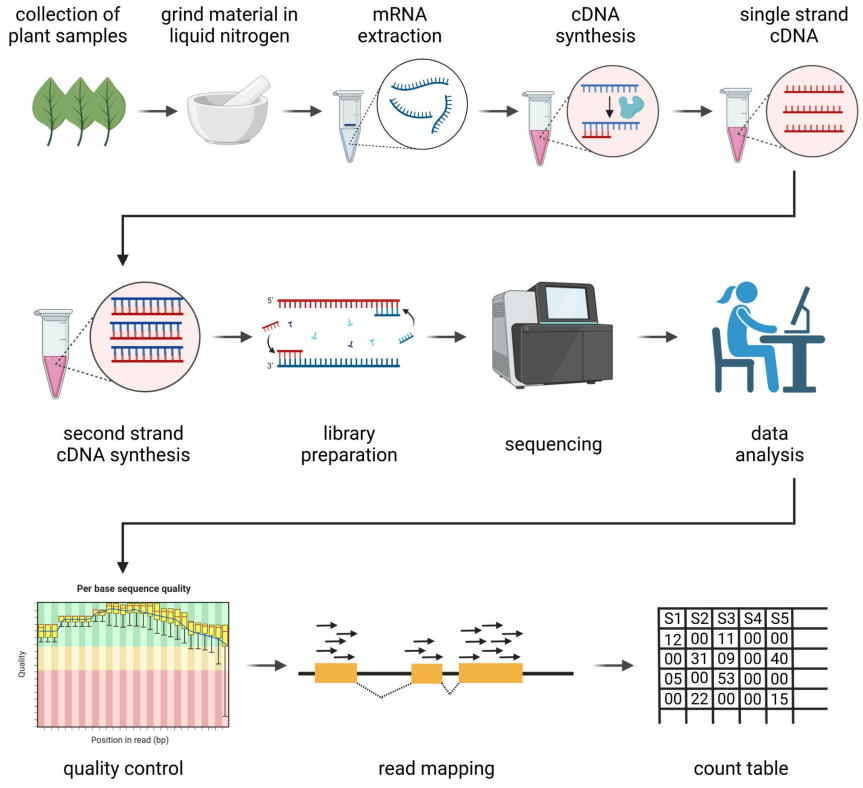
Codon usage

- Protein coding sequences have specific properties
- Codon usage is different between species i.e. different species prefer different codons for certain amino acids
- Rare codons slow down translation (useful between domains)
- Usage of dicodons (=hexamers)
- Codon usage can give additional CDS/ORF support

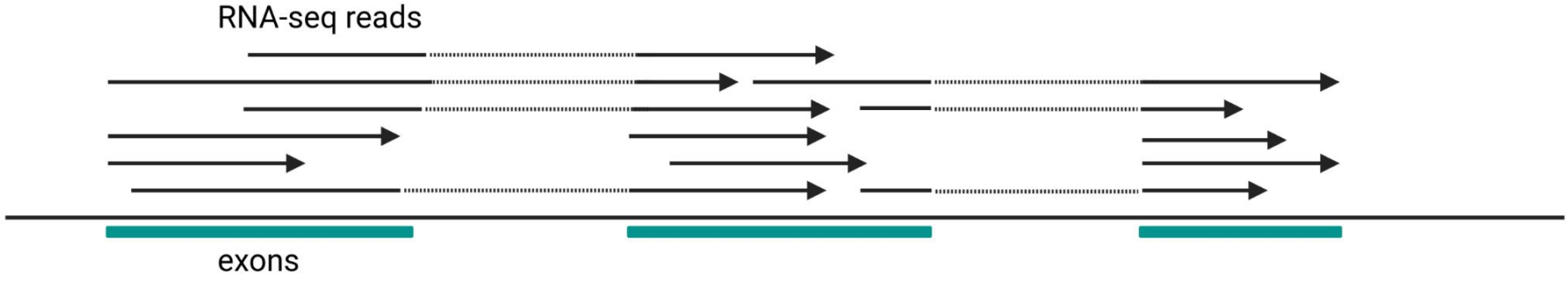


RNA-seq


collection of plant samples
grind material in liquid nitrogen
mRNA extraction
cDNA synthesis
single strand cDNA
second strand cDNA synthesis
library preparation
sequencing
data analysis
Per base sequence quality
Quality
Position in read (bp)
quality control
read mapping
count table
S1 S2 S3 S4 S5
12 00 11 00 00
00 31 09 00 40
05 00 53 00 00
00 22 00 00 15

RNA-seq hints

- Aligned RNA-seq reads indicate exon positions
- Splitting of reads indicates intron positions
- CDS can be identified as ORF within the covered regions



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Annotation quality assessment

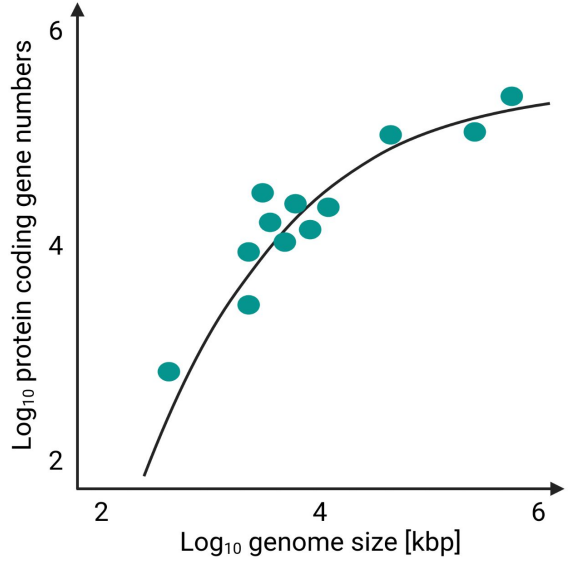
- BUSCO (Benchmarking Universal Single-Copy Orthologs)
  - Example: C:98%[S:85%,D13%],F:1.2%,M0.8%,n:1500
  - C = complete BUSCO genes
  - S = single copy BUSCO genes
  - D = duplicates BUSCO genes
  - F = fragmented BUSCO genes
  - M = missing BUSCO genes
  - n = number of BUSCO query sequences

- NLRome: assessment of high continuity genome sequences

Gene numbers

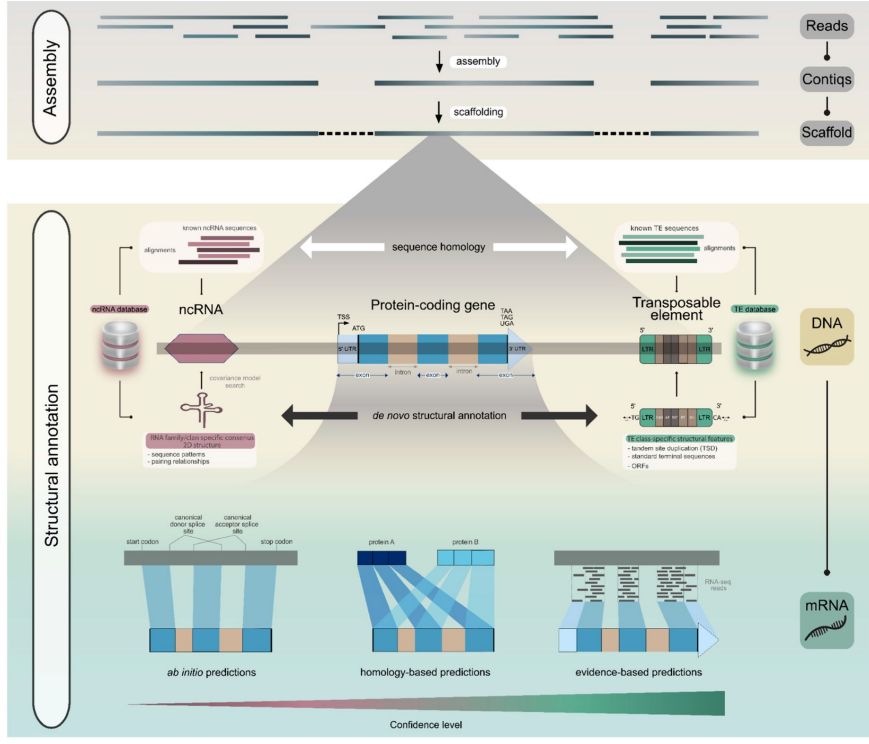
- Average number of genes in plants: 27200

- Gene number is not significantly correlated with genome size



Pucker & Brockington, 2019: 10.1186/s12864-018-5360-z
Michael, 2014: 10.1093/bfgp/elu005

Structural annotation (overview)



de Oliveira et al., 2026: 10.1186/s12864-026-12623-z

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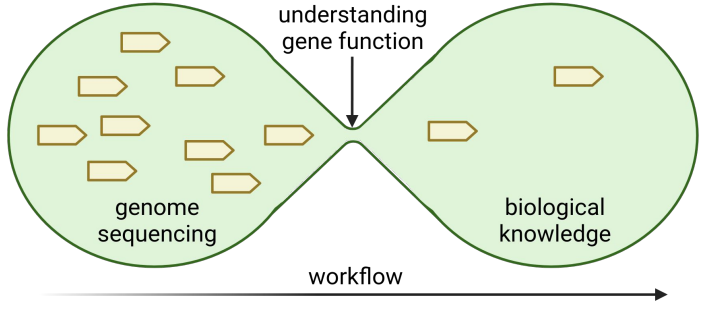
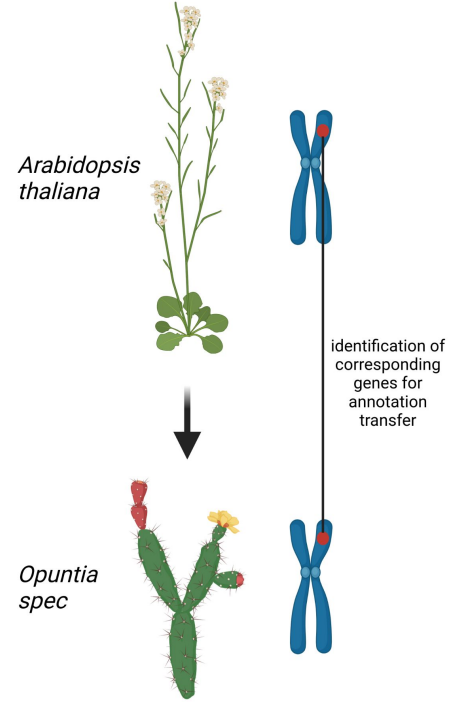
Functional annotation

Functional annotation

- Assumption: similar sequences (genes) have similar functions
- Sequence similarity indicates origin from a shared common ancestor
- Ancestor is likely to have inherited the same function to offspring
- Transfer of functional annotation based on sequence similarity
- *Arabidopsis thaliana* gene functions are well studied and usually serve as reference

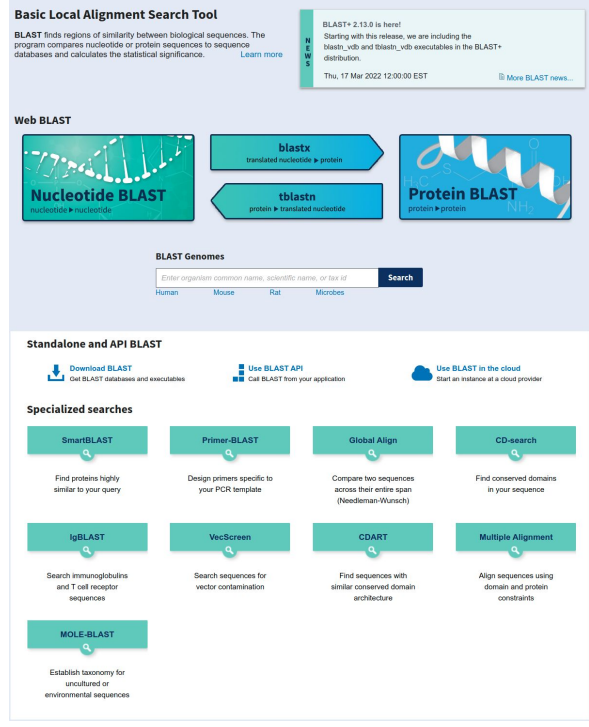




Pucker, 2024: 10.20944/preprints202402.0645.v1

BLAST: Basic Local Alignment Search Tool

- Probably the most famous website of the NCBI
- Comparison of sequences against a large database
- Similar sequences are likely to have similar functions (ideally orthologs)
- Numerous variants of the initial BLASTn were developed



https://blast.ncbi.nlm.nih.gov/Blast.cgi
Altschul et al., 1990: https://doi.org/10.1016/s0022-2836(05)80360-2

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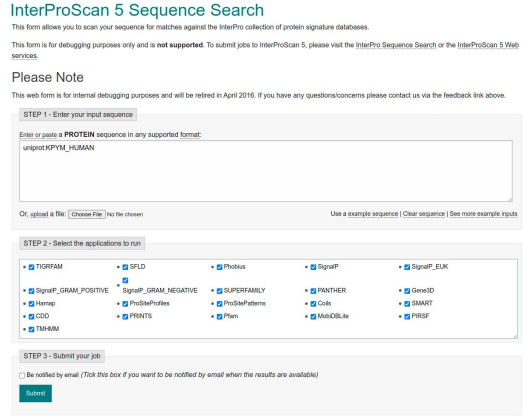
InterProScan5

- Screen of protein sequences against collection of protein signatures
- Allows the assignment of functional annotation terms
- Available as web service, but also as stand alone tool



Results for job iprscan5-I20220320-102557-0980-87120812-p2m

Tool Output | Submission Details

Download in XML format | Download in TSV format | Download in GFF3 format | Download in SVG format | Download HTML tarball file | Download in JSON format

```
sp|P13114|CHSY_ARATH   ba46b0f06bfbf1c91d161f0f91310f43   395   Gene3D   G3DSA:3.40.47.10          -   1     241   4.8E-101      T   20-03-2022
sp|P13114|CHSY_ARATH   ba46b0f06bfbf1c91d161f0f91310f43   395   SUPERFAMILY   SSF53901   Thiolase-like   241   393   2.98E-51      T   20-03-20
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sp|P13114|CHSY_ARATH   ba46b0f06bfbf1c91d161f0f91310f43   395   Pfam   PF02797 Chalcone and stilbene synthases, C-terminal domain   244   394   1.5E-71
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sp|P13114|CHSY_ARATH   ba46b0f06bfbf1c91d161f0f91310f43   395   Gene3D   G3DSA:3.40.47.10          -   242   395   9.7E-62   T   20-03-2022   IPR01603
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sp|P13114|CHSY_ARATH   ba46b0f06bfbf1c91d161f0f91310f43   395   ProSitePatterns   PS00441 Chalcone and stilbene synthases active site.   161   177   -
```


http://www.ebi.ac.uk/Tools/services/web/toolform.ebi?tool=iprscan5&sequence=uniprot:KPYM_HUMAN
Jones et al., 2014: 10.1093/bioinformatics/btu031

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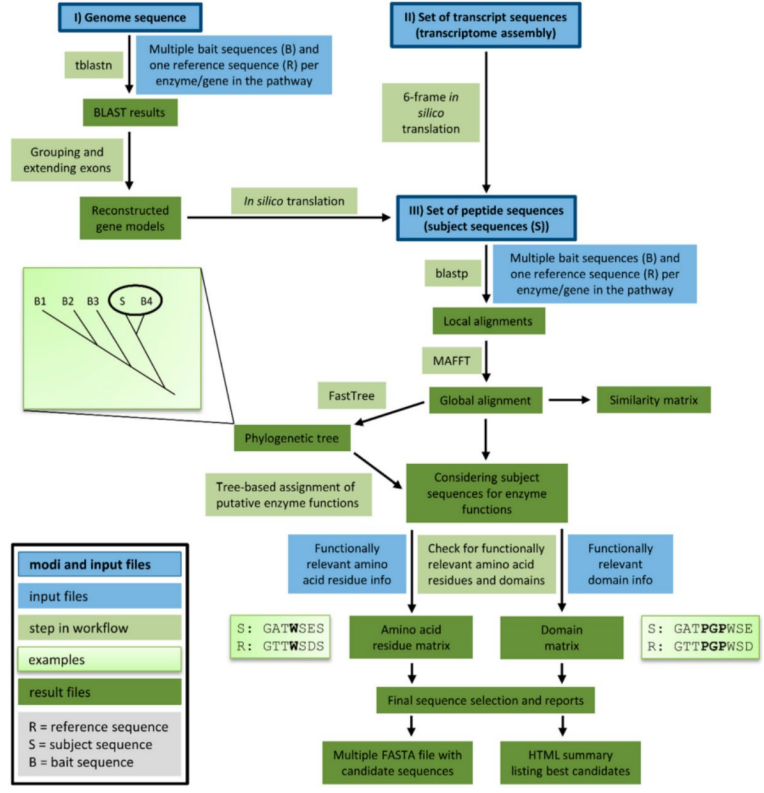
KIPEs

- Identification of all players in a biosynthesis pathway
- Dedicated to the identification of enzymes
- Functionally relevant amino acids need to be known
- Web server: http://www.pbb-tools.de

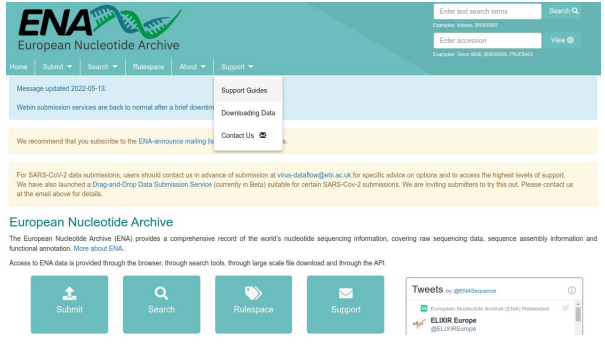


https://github.com/bpucker/KIPEs
Pucker et al., 2020: 10.3390/plants9091103

Databases for submission of sequencing data

- Sequence Read Archive

- European Nucleotide Archive



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Summary

- Sequencing methods: Sanger, Illumina, PacBio, ONT
- Planning the sequencing experiment
- DNA extraction, quality control, library preparation
- Nanopore sequencing & real time monitoring
- Genome sequence assembly
- Structural and functional annotation

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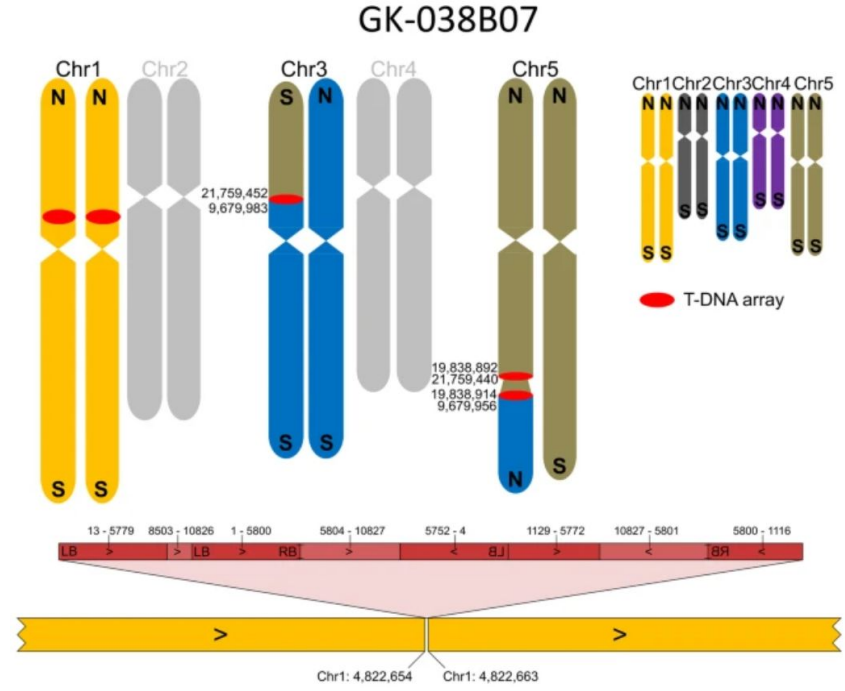
Exercises

Long-read genotyping of T-DNA lines

- T-DNA insertions generate knock-outs
- T-DNA insertion lines were backbone of gene function elucidation in *A. thaliana*
- GABI-Kat comprises >80,000 lines
- Genotyping with ONT sequencing reveals off-targets and number of T-DNAs



Pucker et al., 2021: 10.1186/s12864-021-07877-8

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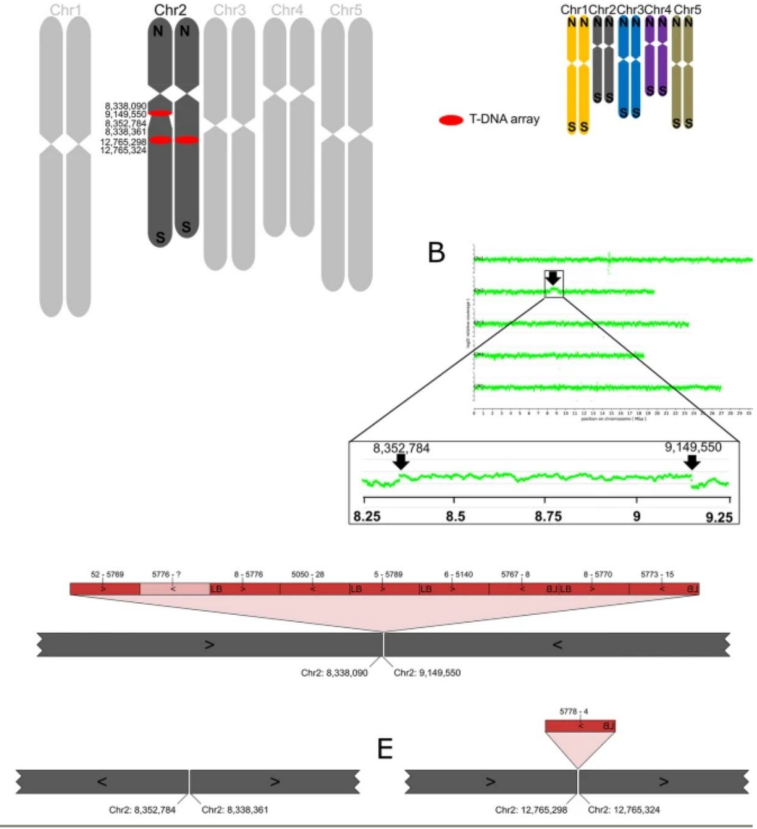
GK-767D12

- Photosynthesis is a fundamental property of plants
- Central genes of the photosynthesis cannot be knocked-out (would be lethal)
- GK-767D12 has a structural variant potentially affecting AT2G21385/CGLD11 (Chloroplast Grana-Less Development 11)



Pucker et al., 2021: 10.1186/s12864-021-07877-8

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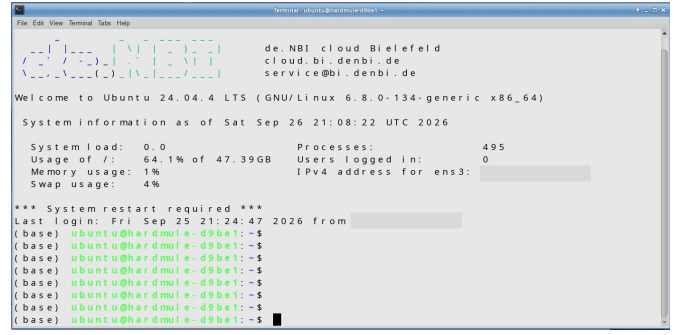
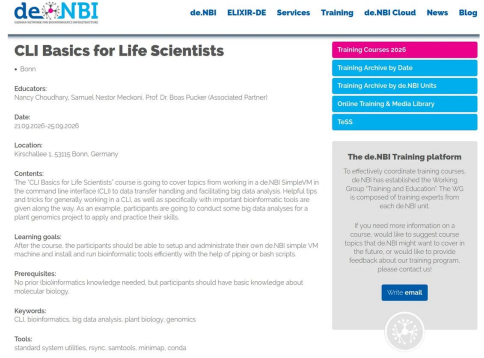
de.NBI

GERMAN NETWORK FOR BIOINFORMATICS INFRASTRUCTURE

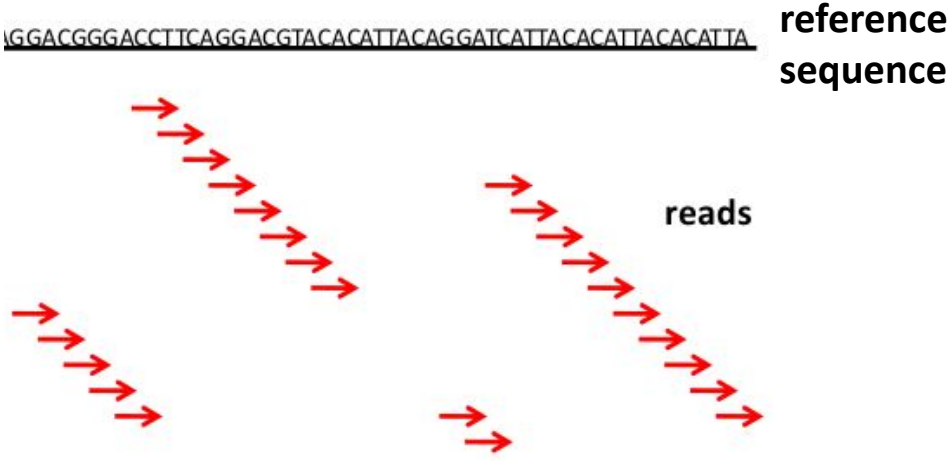
- Cloud computing resources for bioinformatics
- Hosting of services (e.g. pbb-tools.de)
- Training courses for a diversity of methods
- Network for exchange about latest developments in computational biology

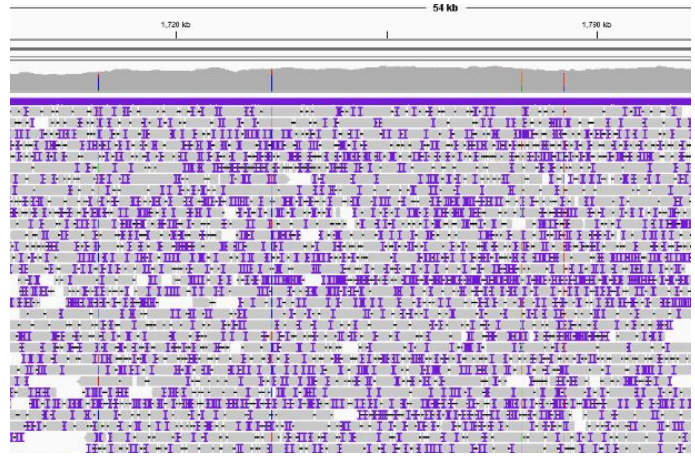




Read mapping: concept & reality

- Alignment of reads against a reference sequence
- SAM = Sequence Alignment/Map
- BAM = Binary Alignment Map





Integrative Genomics Viewer (IGV)

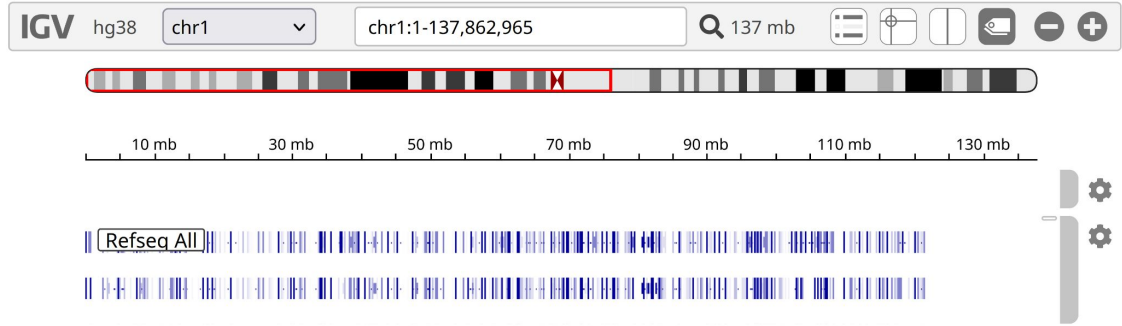
- Download IGV from:

  https://igv.org/doc/desktop/

- or use IGV online:

  https://igv.org/app/

- Load genome from file
- Load file (select BAM file)
- Load file (select GFF file)



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Plan for today

- Connect to a virtual machine (environment for computational analyses)
- Caclulate FASTQ file statistics
- Map reads against a reference genome sequence (TAIR10)
- Download the resulting BAM file
- Inspect your read mapping with IGV

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Starting VM

- Obtain a private key file (matching to public key file in VM)
- Download the README file
- Follow instructions in the README file

https://uni-bonn.sciebo.de/s/cEc5eDsYQPgDRL7

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Exercises

Please find the commands for all exercises here:

https://github.com/bpucker/LRG/blob/main/README_EXERCISES1.md

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Backup: obtaining results

- Download:
  - FASTA + FAI file
  - GFF file
  - BAM + BAI file

- Visualize read mapping in IGV

https://uni-bonn.sciebo.de/s/xb9NXnn4Z8gaTyb

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IGV instructions

- Load genome
- Load from file: GFF, BAM
- Search for gene of interest

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Nanopore sequencing applications

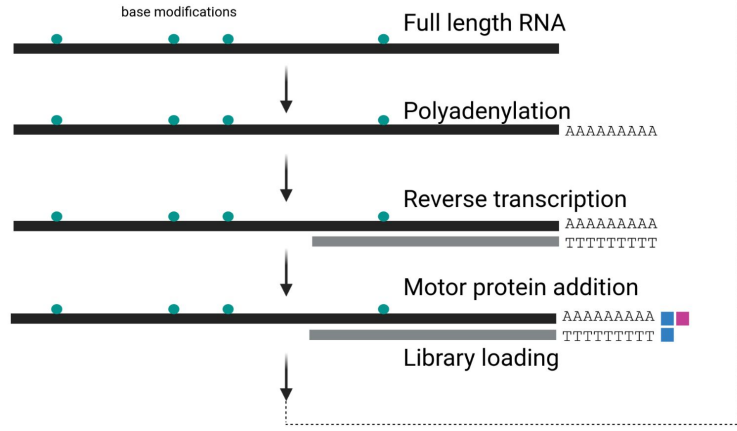
Adaptive sampling

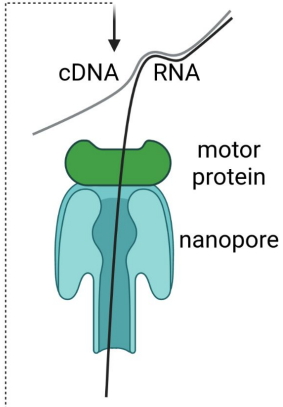
- Check DNA strand based on first sequenced bases against reference sequence:
  - strand is of interest: continue
  - strand is not of interest: reject
  - Example: only sequence bacterial/fungal DNA included in a plant sample

CC BY-SA 2.0, https://commons.wikimedia.org/wiki/File:Maize_crop,_West_Stowell_-_geograph.org.uk_-_1428813.jpg
CC BY 2.5, https://commons.wikimedia.org/wiki/File:Mycorrhizal_root_tips_%28amanita%29.jpg

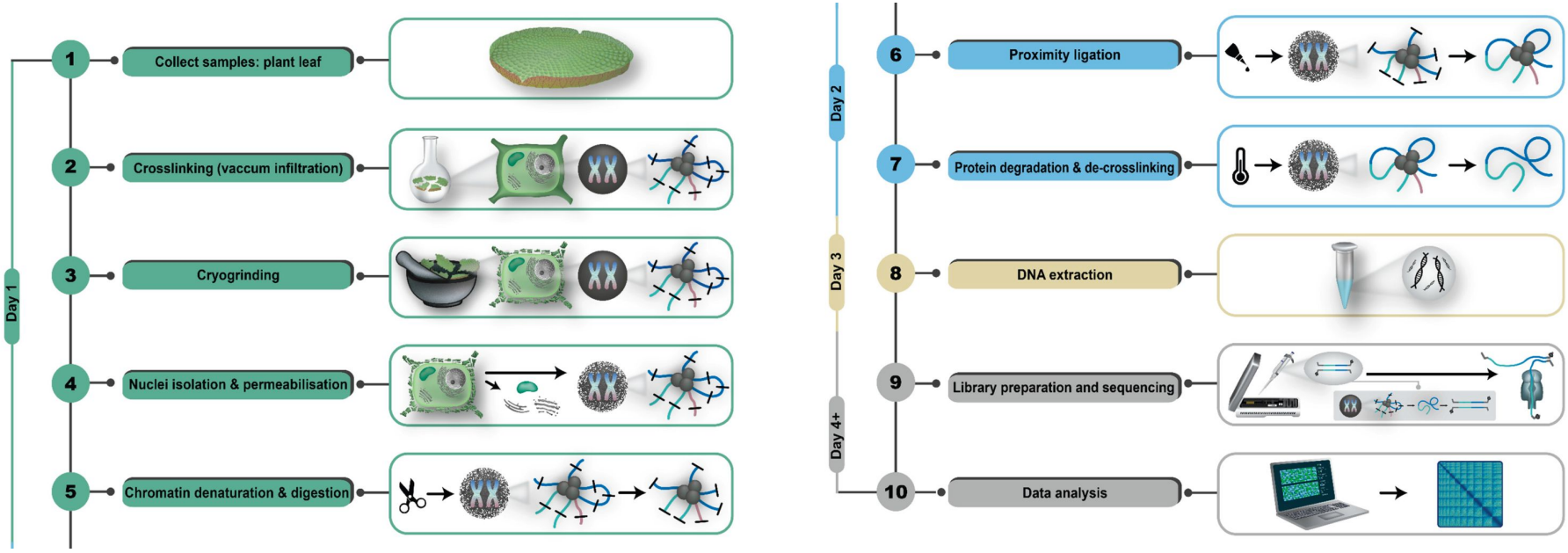
Direct RNA sequencing

- PolyA tail (3') required for primer ligation
- Generation of complementary DNA strand through reverse transcriptase recommended
- Sequencing of RNA strand allows analysis of base modifications





Pore-C (chromatin structure)



de Oliveira et al., 2026: 10.1186/s12864-026-12623-z
https://www.protocols.io/view/pore-c-protocol-for-plant-samples-rm7vz9mmrgx1/v1

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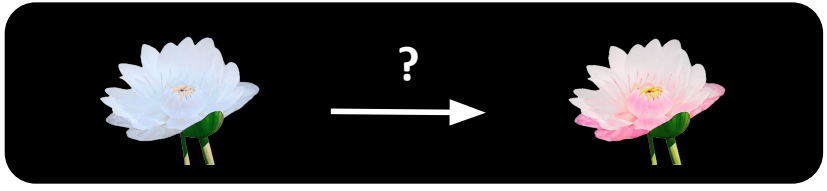
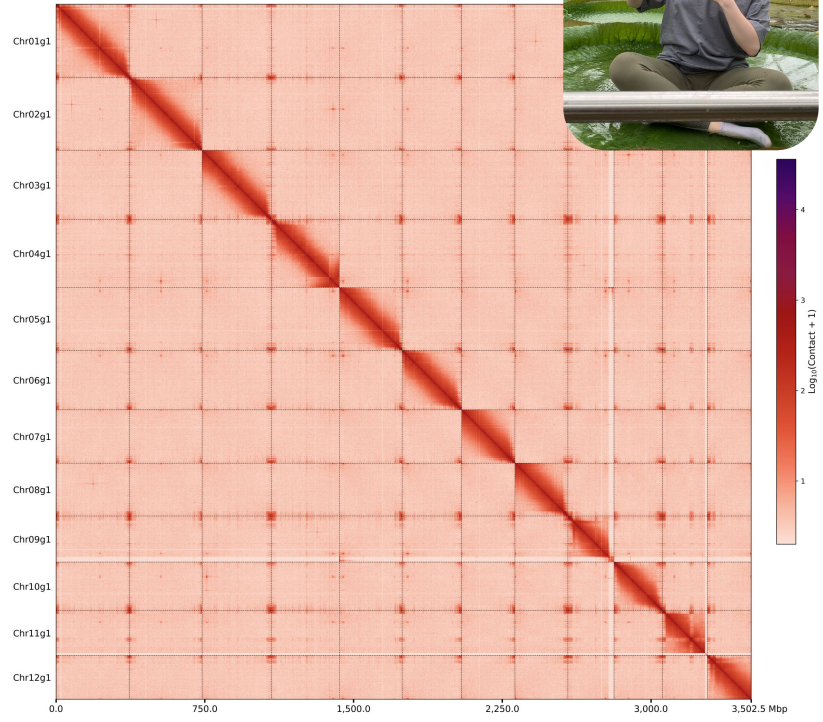
Victoria cruziana

- Water lily with giant floating leaves
- Flowers at night and floral color transitions from white to pink
- Pore-C resulted in an almost perfect genome sequence





Nowak *et al.*, 2025: 10.1101/2024.06.15.599162

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Digitalis purpurea

- Ornamental plant with magenta flowers
- Flower color and shape variation
- Genome sequence reveals underlying genes

Horz *et al.*, 2026: 10.1186/s12864-026-12889-3

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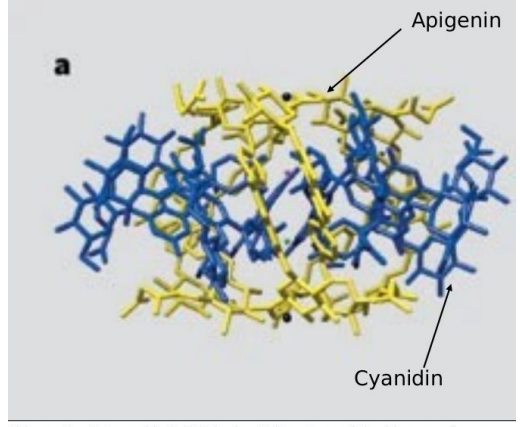
Centaurea cyanus

- Famous for a striking blue flower color
- Complex comprising pigments and metal
- Genome sequence reveals pigment biosynthesis genes



Shiono, M., Matsugaki, N. & Takeda, K. Structure of the blue cornflower pigment. *Nature* **436**, 791 (2005). https://doi.org/10.1038/436791a

SynBio2024 TU Braunschweig
Engineering Blue Flower Colouration

Dassow et al., 2025: 10.1101/2025.01.31.635922

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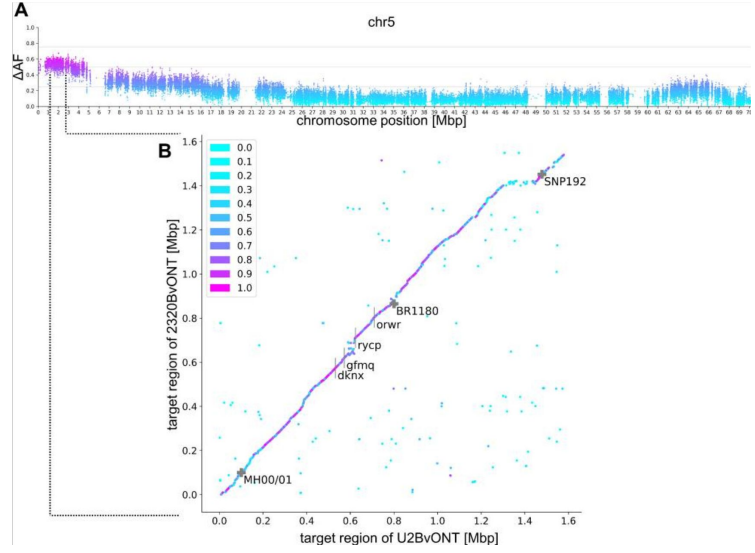
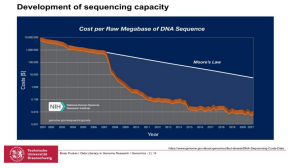
Beta vulgaris

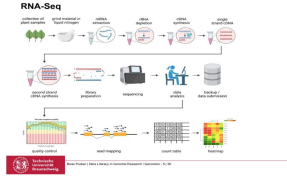
- Important crop in northern hemisphere
- Accounts for about 20% of sugar production
- Nematodes cause serious yield losses

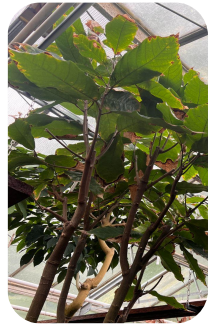


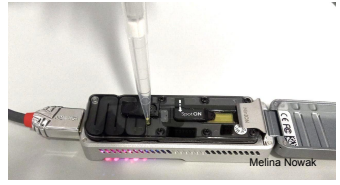
Sielemann et al., 2023: 10.1186/s12864-023-09823-2

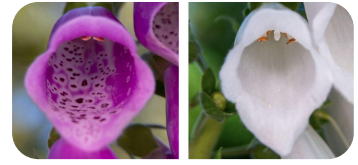
Teaching sequencing in practical courses

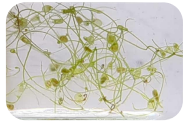












Data Literacy and Plant Genomics

Wolff et al., 2023: 10.1515/jib-2023-0033

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Erasmus+

- Opportunity for exchange within the EU
- Lecturer mobility
- Student mobility e.g. research visit or semester abroad

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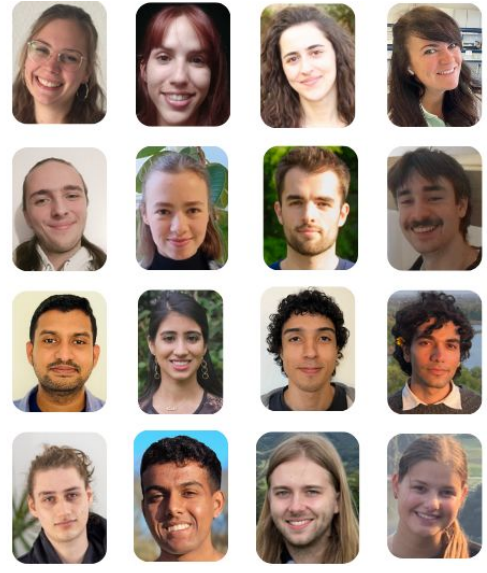
Acknowledgements



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